

Decoding Data Science(DDS) Academy

AI Application Building Challenge

For students (instructor review required)

Day 1 Idea Submission Template

Project Title:

Cookable: An AI-Powered Recipe Assistant

Concept Summary:

(Briefly describe your idea, including its purpose and the problem it aims to solve.)

Cookable is an AI-powered recipe assistant that helps users figure out what to cook using ingredients they already have. You enter what's in your fridge, set your cuisine preference, time available, and dietary restrictions, and the app suggests 5 recipes you can make right now. It also recommends a few ingredients to buy that would unlock even more meal options, with an optional toggle to generate 2 bonus recipes combining your original and new ingredients.

Target Audience:

(Who will benefit from your project? Describe their needs.)

Students and young adults living independently who cook at home but struggle with meal decisions and food waste. They often have random ingredients in the fridge but no clear idea what to make, leading them to order food unnecessarily. They need something fast, practical, and easy to use.

Key Features:

(List the main features or functionalities of your application.)

1. Ingredient input with cuisine, time, and dietary preference filters
2. AI-generated 5 recipe suggestions with steps and prep time using current ingredients

3. Smart ingredient recommendations with reasons, and optional 2 bonus recipes combining old and new ingredients
4. One-click history to revisit and reload any past search instantly

Technical Approach:

(How do you plan to implement your idea? Mention technologies or methods.)

The app will be built using Streamlit for the front end and the Grok API (via the OpenAI-compatible SDK) as the language model powering all recipe and ingredient suggestions. Prompts will be structured to return consistent JSON output which will be parsed and displayed in the interface. The API key will be stored securely using python-dotenv and Streamlit secrets on deployment.

Expected Challenges:

(What potential obstacles do you foresee, and how will you address them?)

- Getting the Grok API to return consistently structured JSON output across different ingredient combinations and filter selections. This will be addressed through careful prompt engineering and iterative testing.
- Handling edge cases such as very few ingredients, highly restrictive dietary filters, or unusual combinations without breaking the app. This will be handled through input validation and fallback prompts.

Submission Format:

(Outline how your submission should be presented: documents, slides, etc.)

- Deployed Streamlit app (live link)
- GitHub repository with full source code and README
- Demo video showing the full user flow
- Daily LinkedIn posts documenting progress throughout the challenge

Expected Outcome:

(What do you aim to achieve by the end of the challenge?)

A fully deployed, working AI recipe assistant that real users can open in a browser, enter their fridge ingredients, and get actionable meal ideas within seconds. The app should be stable, handle edge cases gracefully, and produce a clean PDF output. By the end of the challenge, Cookable should be portfolio-ready and genuinely useful beyond the challenge itself.

Additional Notes (Optional):

The "Get These" ingredient suggestion feature is optional for the user to act on -- the app surfaces the recommendations automatically but does not require the user to follow them. The bonus recipe section only appears if the user opts in, keeping the experience clean and uncluttered.